THE FILTERED-COLIMIT EXTENSION OF AN ADDITIVE FUNCTOR IS ADDITIVE AND COMMUTES WITH ARBITRARY DIRECT SUMS

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file functors.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 0 hits for functors.tex (fu-occupied.json, 2026-10-14); consistent with the duty manager's six-check dispatch.

Source reference. Original lemma 0GNJ: sources/stacks-chow/tags.txt, source lines 20794–20794. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0DLP: sources/stacks-chow/tags.txt, source lines 17054–17054. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let A, $\mathcal{B}$, F be as in Lemma 0GNJ. Assume $\mathcal{B}$ is additive and F is additive. Then F' is additive and commutes with arbitrary direct sums.*

Proof. To show that F' is additive it suffices to show that $F'(M) \oplus F'(M') \to F'(M \oplus M')$ is an isomorphism for any A-modules M, M', see Homology, Lemma 0DLP. Write $M = \operatorname{colim}_i M_i$ and $M' = \operatorname{colim}_j M'_j$ as filtered colimits of finitely presented A-modules M_i. Then $F'(M) = \operatorname{colim}_i F(M_i)$, $F'(M') = \operatorname{colim}_j F(M'_j)$, and

$$\begin{aligned} F'(M \oplus M') &= F'(\operatorname{colim}_{i,j} M_i \oplus M'_j) \\ &= \operatorname{colim}_{i,j} F(M_i \oplus M'_j) \\ &= \operatorname{colim}_{i,j} F(M_i) \oplus F(M'_j) \\ &= F'(M) \oplus F'(M') \end{aligned}$$

as desired. To show that F' commutes with direct sums, assume we have $M = \bigoplus_{i \in I} M_i$. Then $M = \operatorname{colim}_{I' \subset I \text{ finite}} \bigoplus_{i \in I'} M_i$ is a filtered colimit. We obtain

$$\begin{aligned} F'(M) &= \operatorname{colim}_{I' \subset I \text{ finite}} F'(\bigoplus\nolimits_{i \in I'} M_i) \\ &= \operatorname{colim}_{I' \subset I \text{ finite}} \bigoplus\nolimits_{i \in I'} F'(M_i) \\ &= \bigoplus\nolimits_{i \in I} F'(M_i) \end{aligned}$$

Date: stacks-project:functors.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

The second equality holds by the additivity of F' already shown. $\square$